Sardar Vallabhbhai National Institute of Technology, Surat
Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401) — Laboratory Manual
B.Tech IV (AI) — Semester VII

Experiment 2: Production Rules and the Match Phase

Aim

To represent domain knowledge as IF–THEN production rules, hold facts in a working memory, and compute the conflict set of applicable rules.

Theory

A production system has three parts: a rule base of IF–THEN productions, a working memory (WM) of currently believed facts, and an inference engine that runs the recognise–act cycle: match all rules against WM, select one rule from the resulting conflict set, act by executing its conclusion, and repeat.

This experiment isolates the match phase. A rule is applicable when every condition in its IF part is present in WM and its conclusion is not already there (refractoriness — preventing a rule from firing repeatedly on the same data). The set of applicable rules is the conflict set; choosing among them is the subject of Experiment 5.

Algorithm

1. Encode each rule as a record with an identifier, a list of conditions and a conclusion.
2. Initialise working memory as a set of facts.
3. For each rule, test whether all its conditions are members of WM and its conclusion is absent.
4. Collect and display the conflict set.

Exercises

1. Add facts to WM one at a time and record how the conflict set grows or shrinks.
2. Extend the matcher to support negated conditions of the form 'NOT fact'.
3. Measure (with time.perf_counter) how match time grows as you scale to 1,000 auto-generated rules, and relate the result to why the RETE algorithm exists.

Viva Questions

1. Describe the recognise–act cycle.
2. What is refractoriness and why is it needed?
3. What problem does the RETE matching algorithm solve?